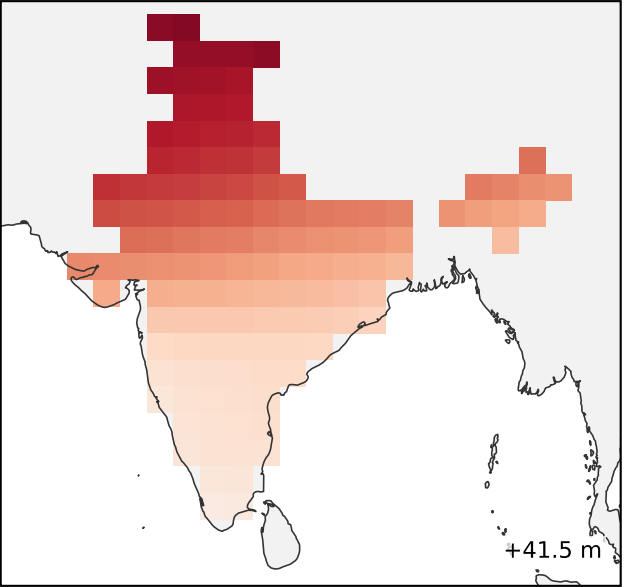


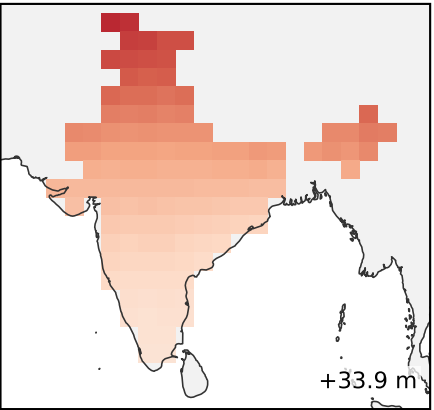
Case study 2 — Z500: init 2026-03-19, verified at week-6 lead (India only)

Week-6 mean anomaly (forecast days 36–42) — India land pixels only (value = All-India area mean)

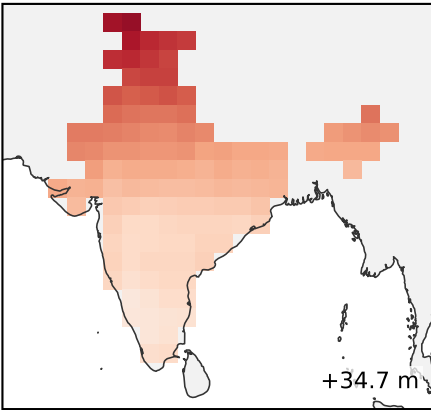
ERA5 (obs)



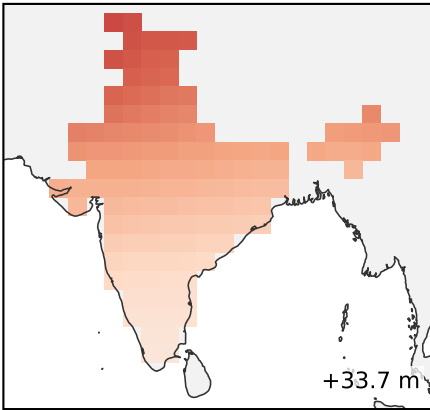
SPIRE



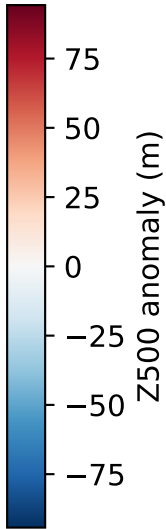
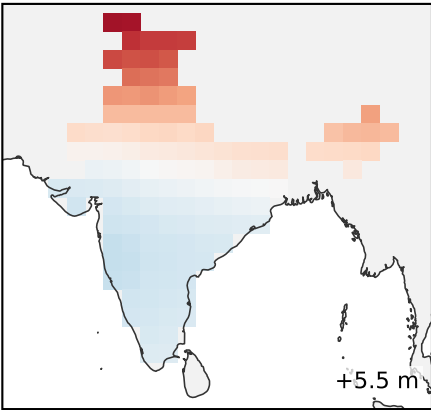
FuXi-S2S



ECMWF



NCEP



How to read:

Black line = ERA5 observed
Colour line = Model forecast
Shaded band = $\pm 1\sigma$ ensemble
Grey area = Week-6 window
Dashed zero = climatology

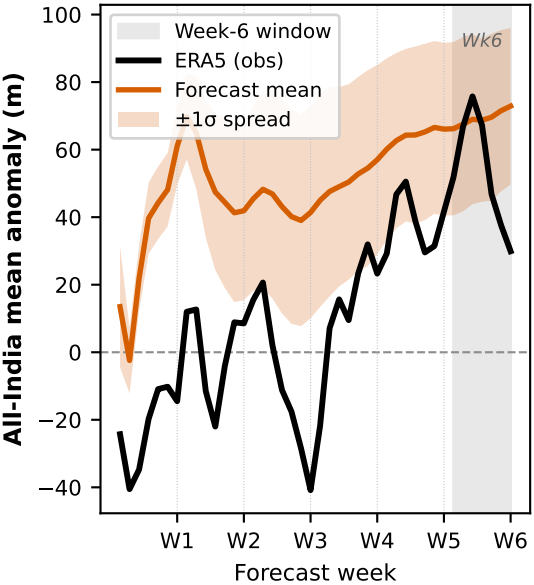
Key Finding

ERA5 shows a strong positive Z500 anomaly (warm ridge, NW India) at week-6.

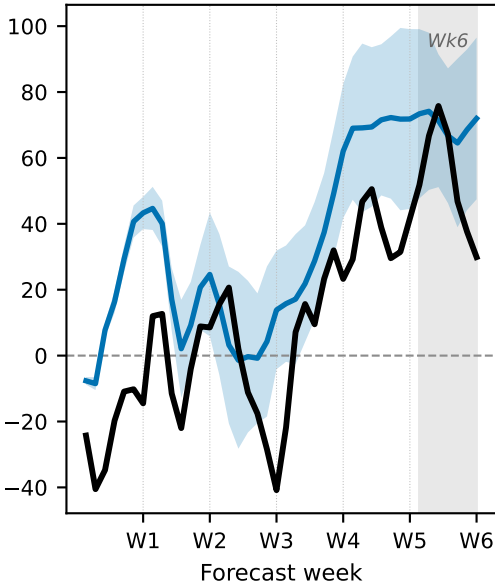
FuXi-S2S predicts the opposite sign (cold trough) — a systematic dynamical bias at long leads.

SPIRE/ECMWF/NCEP get the sign but strongly underestimate amplitude.

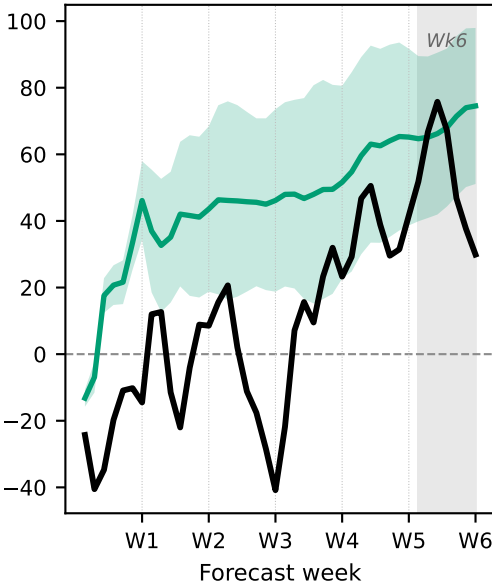
SPIRE



FuXi-S2S



ECMWF



NCEP

